

Section 9: Model Context Protocol (MCP)

The USB-C for AI Applications

9.1. Understanding MCP



Limitations

Generative AI development has challenges because every provider, tool, and model has its own way of working. This makes building and scaling solutions slow and complex.

- **Integration issue** → The **MxN** problems
- **No common standard** → every integration needs custom work & extra dev effort
- **Security gaps** → sharing data safely across systems is tricky

MCP

The Model Context Protocol (MCP) is an open-source standard that provides a unified, plug-and-play way for AI models, particularly large language models (LLMs), to connect with and use external tools, data sources, and services. (by Anthropic)

- **Universal connector** → acts like USB-C for all AI tools and resources
- **Easier scaling** → removes integration hurdles and speeds up development
- **Data Access** → Controlled access to resources and contextual data

MCP - Core Components



MCP Host

The environment where the LLM runs (like an IDE or AI app). It's the control center that connects users to tools and data through MCP.

MCP Client

The bridge between the host and servers. It handles all communication, turning requests and responses into a common MCP format.



MCP Server

The actual tool or data source exposed via MCP. It provides functions or information to the host in a standardized, easy-to-use way.

MCP - Flow

